(11) Faraday's law:

$$E = -\frac{d\Phi_B}{dt}$$

$$\mathcal{E} = -N\frac{d\Phi_B}{dt}$$

(111) Self Inductance: (L)

(111)



$$\Phi \propto i$$

$$\Rightarrow \Phi = Li$$

Here, L = self inductance

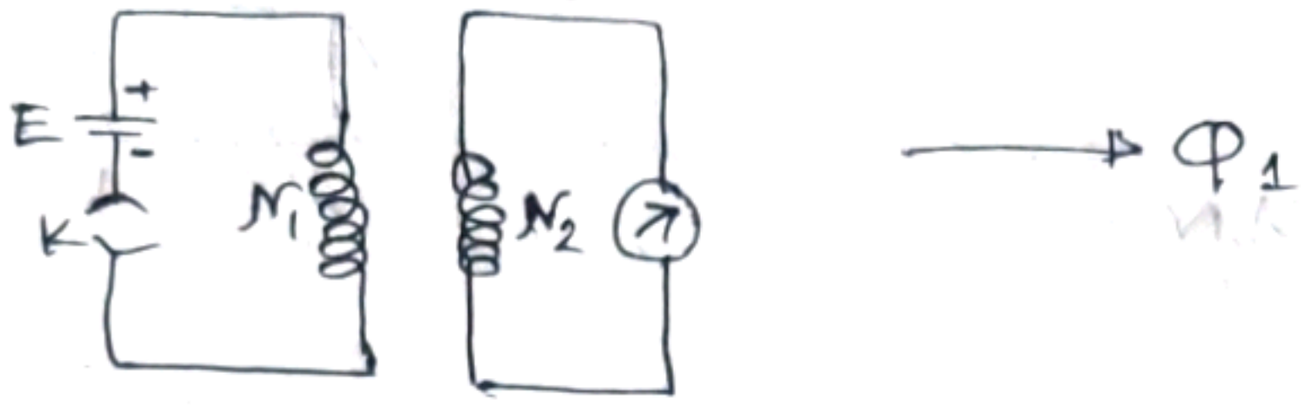
Again, $E = -\frac{d\Phi}{dt}$

$$= -\frac{d}{dt}(Li)$$

$$= -L\frac{di}{dt}$$

$$\Rightarrow L = -\frac{E}{\frac{di}{dt}}$$

(iv) Mutual Inductance:



Here, $\Phi_1 \propto i$

$\Rightarrow \Phi_1 = Mi$

Here, M = Mutual Inductance

Again, $E = -\frac{d\Phi_1}{dt}$

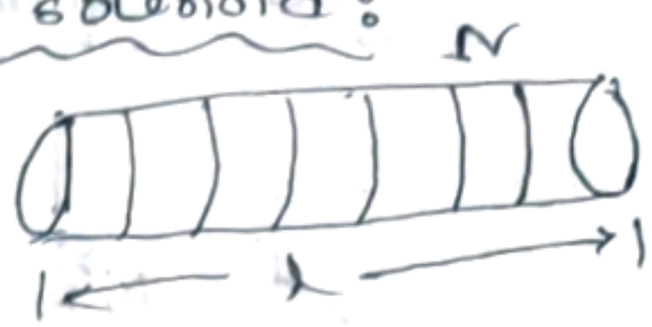
$= -\frac{d}{dt}(Mi)$

$= -M\frac{di}{dt}$

$\Rightarrow M = -\frac{E}{\frac{di}{dt}}$

$$\therefore M = -\frac{E}{di/dt}$$

(v) Self Inductance for a long solenoid:



We know,

$$B = \frac{\mu_0 N I}{l}$$

$$\varphi_B = BA$$

$$= \frac{\mu_0 N I}{l} \cdot A$$

$$= \frac{\mu_0 N I A}{l}$$

Number of turns $= N$, Then

$$\varphi_B = \frac{\mu_0 N I A}{l} \cdot N$$

$$= \frac{\mu_0 N^2 I A}{l}$$

Again, $\varphi = LI$

$$\Rightarrow L = \frac{\varphi}{I} = \frac{\mu_0 N^2 I A}{l} \times \frac{1}{I}$$

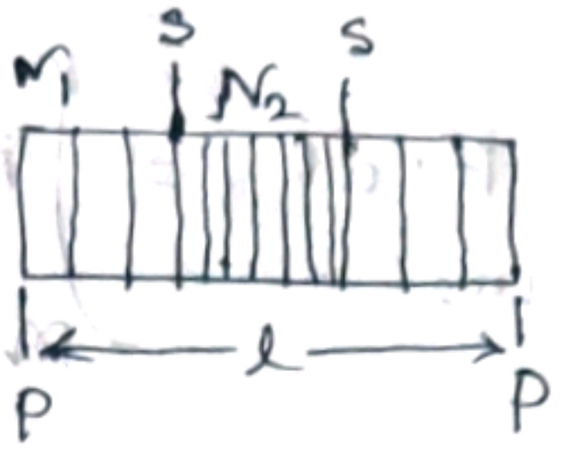
$$\therefore \boxed{L = \frac{\mu_0 N^2 A}{l}}$$

If permitivity $= \mu_r$ add, then,

$$L = \frac{\mu_0 N^2 A}{l} \cdot \mu_r$$

$$\therefore \boxed{L = \frac{\mu N^2 A}{l}} \quad (\because \mu_0 \times \mu_r = \mu)$$

(vi) Mutual inductance for two coaxial cell:



For P,

$$B = \frac{\mu_0 N_1 I}{l}$$

$$\therefore \phi = \frac{\mu_0 N_1 I A}{l}$$

when N_2 also add then,

$$\phi = \frac{\mu_0 N_1 N_2 I A}{l}$$

Again, $\phi = MI$

$$\Rightarrow M = \frac{\phi}{I} = \frac{\mu_0 N_1 N_2 I A}{l} \times \frac{1}{I}$$

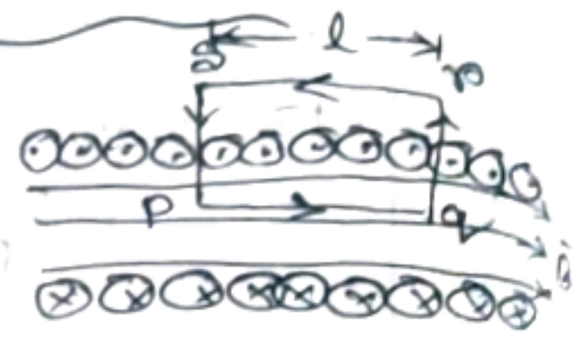
$$= \frac{\mu_0 N_1 N_2 A}{l}$$

If permittivity μ_r,

Then $M = \frac{\mu_0 \mu_r N_1 N_2 A}{l}$

$$= \frac{\mu N_1 N_2 A}{l}$$

Megnetic field inside a long solenoid :



$$\oint_{pqrs} \bar{B}\cdot d\bar{l} = \int_{pq} \bar{B}\cdot d\bar{l} + \int_{qr} \bar{B}\cdot d\bar{l} + \int_{rs} \bar{B}\cdot d\bar{l} + \int_{sp} \bar{B}\cdot d\bar{l} \quad \text{—(1)}$$

pq and rs are in alternative each other.

$$\therefore \int_{pq} \bar{B}\cdot d\bar{l} = \int B\,dl = Bl$$

$$\text{and} \int_{qr} \bar{B}\cdot d\bar{l} = \int_{sp} \bar{B}\cdot dl = 0$$

$$\text{and} \int_{sr} \bar{B}\cdot d\bar{l} = 0 \qquad [\because \bar{B} = 0]$$

From equ^{n} (1), we get,

$$\oint \bar{B}\cdot d\bar{l} = \int B\,dl$$

$$= Bl$$

From Amperes Law, $\oint \bar{B}\cdot d\bar{l} = \mu_0 \times$ net current enclosed by the path.

$$\Rightarrow Bl = \mu_0 (nl) \times i$$

$$\boxed{\therefore B = \mu_0 n i}$$

Date: 17.01.22 lecture: 18

State and explain Amperes Law:



$\Rightarrow \oint B \cdot dl = \mu_0 i$

statement:

Explaination:

We know, $\oint \bar{B} \cdot d\bar{l} = \oint B \, dl$

$= B \oint dl$

$\therefore \oint \bar{B} \cdot d\bar{l} = B \times 2\pi r$ ——— (1)

Again, $B \propto i$ and $B \propto \frac{1}{r}$

So, $B \propto \frac{i}{r}$

$\Rightarrow B = \frac{\mu_0}{2\pi} \cdot \frac{i}{r}$

Putting the value of B in equ (1), we get,

$\oint \bar{B} \cdot d\bar{l} = B \times 2\pi r$

$= \frac{\mu_0 i}{2\pi r} \times 2\pi r$

Therefore, $\oint \bar{B} \cdot d\bar{l} = \mu_0 i$ (Hence proved)

Define: Motional emf.

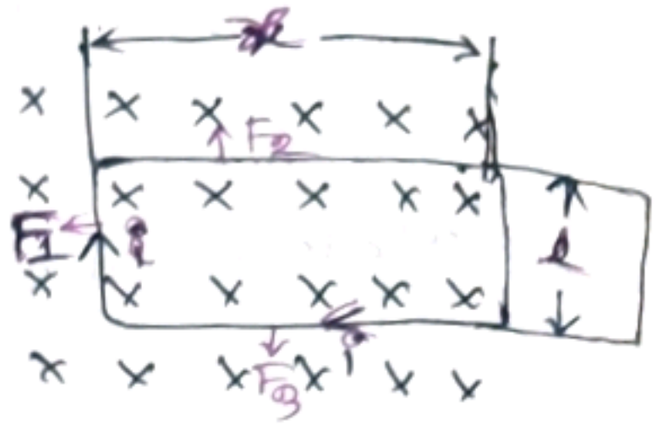
Prove that, Motional emf, $P = \frac{B^2 l^2 v^2}{R}$

Soln:

We know.

$$\Phi_B = BA$$

$$= Blx \quad \text{(Here, } A = lx\text{)}$$



According to faradays law,

$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

$$= -\frac{d}{dt}(Blx)$$

$$= -Bl\frac{dx}{dt}$$

$$\therefore \mathcal{E} = Blv \quad \left[v = -\frac{dx}{dt} \text{ in the induced emf}\right]$$

Again, $i = \frac{\mathcal{E}}{R} = \frac{Blv}{R}$

Here, $F_2 + F_3 = 0$

and $F_1 = il \times B$

$$= ilB\sin\theta$$

$$= \frac{Blv}{R} \cdot Bl \times \sin 90^\circ \quad \left[\because i = \frac{Blv}{R} \text{ and } \theta = 90^\circ\right]$$

$$\Rightarrow F_1 = \frac{B^2 l^2 v}{R}$$

Again,

We know,

Motional emf, $P = F_1 v$

$= F_1 v$

$= \frac{B^2 l^2 v}{R} \times R$

Therefore, $P = \frac{B^2 l^2 v^2}{R}$

(Hence proved)

☞ A solenoid having an air core and 10 cm long has 100 turns and its area of cross-section is 5 sqcm. Find the co-efficient of self inductance of the solenoid.

Soln:

We know,

self inductance,

$$L = \frac{\mu_0 N^2 A}{l}$$

$$= \frac{4\pi \times 10^{-7} \times (100)^2 \times 5 \times 10^{-4}}{10 \times 10^{-2}}$$

$\therefore L = 6.28 \times 10^{-5}$ Henry Ans

Here,

$l = 10$ cm

$= 10 \times 10^{-2}$ m

$N = 100$

$A = 5$ sq cm

$= 5 \times 10^{-4}$ m^2

$\mu_0 = 4\pi \times 10^{-7}$

☞ Calculate the self inductance of a solenoid having 1000 turns and length 1m. The area of cross-section is $7cm^2$ and relative permeability of the core is 1000.

Solⁿ

We know,

$$L = \frac{\mu_0 \mu_r A N^2}{l}$$

$$= \frac{4\pi \times 10^{-7} \times 1000 \times 7 \times 10^{-4}}{1m}$$

$$= 87.96 \times 10^{-7}$$

$$\therefore L = 8.79 \times 10^{-5} \text{ Henry}$$

Ans.

Here,

$\mu_0 = 4\pi \times 10^{-7}$

$\mu_r = 1000$

$A = 7cm^2$

$= 7 \times 10^{-4} m^2$

$N = 1000$

$= 10^{+3}$

$\Rightarrow N^2 = 10^6$

$l = 1$

Lecture: 19

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Question: A solenoid of length 30 cm and area of cross-section 10 sqcm has 1000 turns wound over a core of constant permeability 600. Another coil of 500 turns is wound the same coil at its middle.

Calculate the mutual inductance.

Solⁿ:

We know,

$$M = \frac{\mu_0 \mu_r N_1 N_2 A}{l}$$

$$= \frac{4\pi \times 10^{-7} \times 600 \times 10^{+3} \times 5 \times 10^{+2} \times 10 \times 10^{-4}}{30 \times 10^{-2}}$$

$$= \frac{4\pi \times 10^{-11} \times 10 \times 6 \times 10^{7} \times 5}{30 \times 10^{-2}}$$

$$= 4\pi \times 10^{-11+8+2}$$

$$= 1.2577 \times 10^{-1+1}$$

$\therefore M = 1.257$ Henory. Ans.

Here,

$\mu_0 = 4\pi \times 10^{-7}$

$\mu_r = 600$

$N_1 = 1000$

$N_2 = 500$

$A = 10$ sq cm

$= 10 \times 10^{-4}$ sq m

$l = 30$ cm

$= 30 \times 10^{-2}$ m

Question: Two coils a primary of 600 turns and a secondary of 30 turns are wound on an iron ring of mean radius 0.1m and cross section 4×10^{-4} m. Find mutual inductance (μ_r for iron = 800).

Solⁿ:

We know,

$$M = \frac{N_1 N_2 \mu_0 \mu_r A}{l}$$

$$= \frac{600 \times 30 \times 4\pi \times 10^{-7} \times 800 \times 4\pi \times 10^{-4}}{0.2\pi}$$

$$= 3.62 \times 10^{9-8-11}$$

$\therefore M = 3.62 \times 10^{-2}$ Henory Ans.

Given,

$N_1 = 600$

$N_2 = 30$

$\mu_0 = 4\pi \times 10^{-7}$

$\mu_r = 800$

$A = \pi r^2$

$= \pi (4 \times 10^{-2})^2$

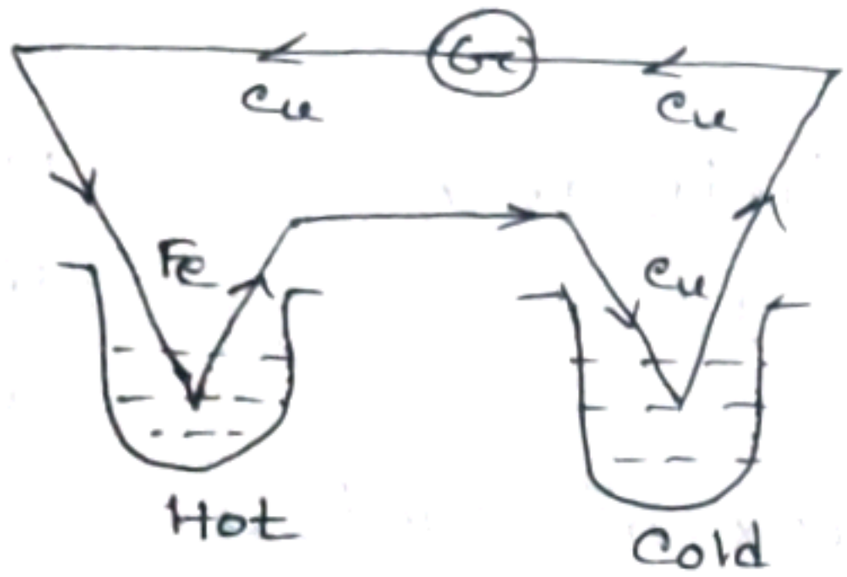
$= 4\pi \times 10^{-4}$

$l = 2\pi r$

$= 2\pi \times (0.1)$

$= 0.2\pi$

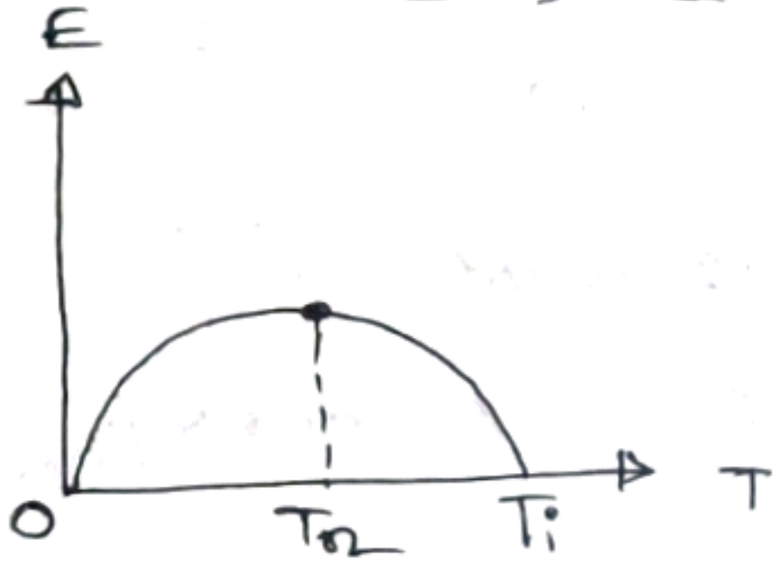
Thermo - Electricity



☞ Seebeck effect → This pp effect

☞ Thermo-electricity → This process

☞ Thermo-couple : Fe & Cu



☞ Neutral Temperature (T_n) [when E_{max}]

☞ Temperature of inversion (T_i) [when $E = 0$]

We know,

$$E = at + bt^2 \quad \text{---(i)}$$

$$\Rightarrow \frac{dE}{dt} = a + 2bt \quad \text{---(ii)}$$

when $t = t_n$, then $E = E_{max}$,

and $\frac{dE}{dt} = 0$

Now
putting $\frac{dE}{dt} = 0$ in eqⁿ (II), we get

$$a + 2bt_n = 0$$

$$\Rightarrow t_n = -\frac{a}{2b} \quad \text{——— (III)}$$

Again, For inversion temperature,

$$t = t_i \quad \text{and} \quad E = 0.$$

Now, putting $E = 0$ in eqⁿ (I), we get,

$$0 = at_i + bt_i^2$$

$$\Rightarrow t_i (a + bt_i) = 0$$

$$\Rightarrow a + bt_i = 0$$

$$\therefore t_i = -\frac{a}{b} \quad \text{——— (IV)}$$

comparing (III) and (IV),

we get, $\boxed{t_i = 2t_n}$.

— × —

Question:

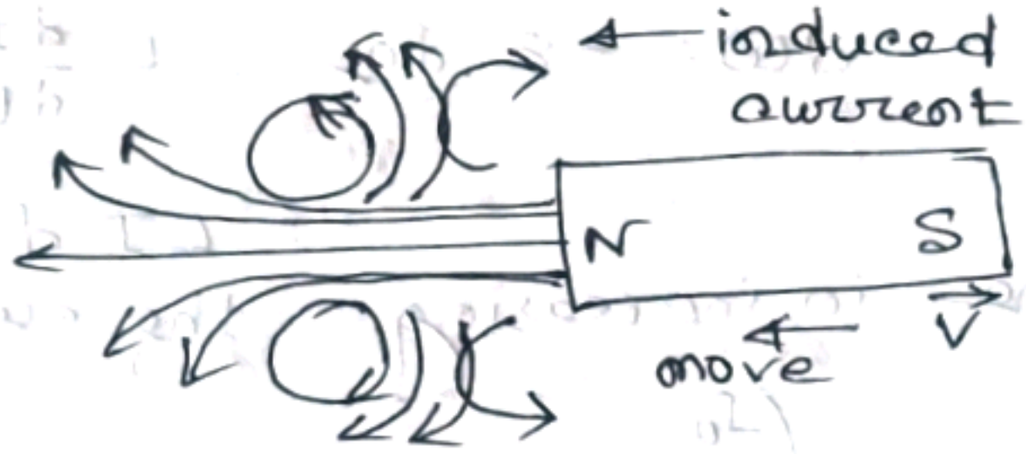
Laws of thermo emf.
- (I) Law of Intermediate metal
- (II) Law of Intermediate Temperature

$$E_1^3 = E_1^2 + E_2^3$$

☞ Lenz's law states that, "The flux of the magnetic field due to the induced current opposes the change in the flux that cause the induced current.

Explanation:

If the change of in the flux is increases then ~~th~~ the lenz's law required that the direction of the induced current oppose the increases. This means that flux of the magnetic field of the induced current would pass through in the loop in a direction opposite to that of the original flux that increasing in fig (a).



Original flux.

② Consider an electric circuit containing inductance. When the circuit is closed, a back-emf is induced in the circuit which opposes the growth of current in it.

let. I be the current and the inductance L at any instant t.

let. the rate of the growth of $\overset{\text{The current}}{\text{back-emf}} = \frac{dI}{dt}$

The back-emf induced in the inductor,

$$\mathcal{E} = -L\frac{dI}{dt}$$

The work done in moving charge dq,

$$dW = -\mathcal{E}\,dq = L\frac{dI}{dt}\cdot dq = L\frac{dq}{dt}dI$$

$$= LI\,dI$$

Total work done in increasing the current from 0 to I_0,

$$W = \int_0^{I_0} LI\,dI = \frac{1}{2}LI_0^2$$

Thus the energy stored in the inductor is

$$U = \frac{1}{2}LI_0^2$$

For a solenoid,

magnetic induction, $B = \frac{\mu_0 NI}{l}$

$$\Rightarrow I = \frac{Bl}{\mu_0 N}$$

and self conductance, $L = \frac{\mu_0 N^2 A}{l}$

Now,

Energy stored in a magnetic field,

$$U = \frac{1}{2} L I^2$$

$$= \frac{1}{2} \cdot \frac{\mu_0 N^2 A}{l} \cdot \frac{Bl}{\mu_0 I} \left(\frac{Bl}{\mu_0 N}\right)^2$$

$$= \frac{1}{2} \frac{\mu_0 N^2 A B^2 l^2}{l \mu_0^2 N^2}$$

$$\therefore U = \frac{1}{2} \frac{B^2 l A}{\mu_0}$$

Q5 $E = at + bt^2$

Now, $t°c = (T - 273)°C$ where T is the absolute degrees.

$$\therefore E = a(T-273) + b(T-273)^2$$

$$\rightarrow \frac{dE}{dT_n} = a + 2b(T - 273)$$

$$\rightarrow \frac{d^2E}{dT^2} = 2b$$

(1) $T = T_n$, $\frac{dE}{dT} = 0$

$$\therefore a + 2b(T_n - 273) = 0$$

$T_n =$ ——————

(11) T_i be ——— T of. g_a - - -

$$T_n = \frac{(273 + T_i)}{2}$$

T_i

$$0 = a + 2b\,(T_n - 273)$$

$$\Rightarrow T_n - 273 = -\frac{a}{2b}$$

$$\Rightarrow T_n = \left(273 - \frac{a}{2b}\right) K$$

$$= -\left(\frac{a}{2b}\right)^{\circ}C$$

$$T_i = 2T_{n2} - 273$$

$$= 2\left(273 - \frac{a}{2b}\right) - 273$$

$$= 273 - \frac{a}{b} \quad K$$

$$= -\left(\frac{a}{b}\right)^{\circ}C$$

~~$t = (273 - T)$~~

$t = T - 273$

$T = t + 273$

(111) $\gamma = T\,\frac{dE}{dT} = (t + 273)(a + 2bt)$

(IV) $C = T\,\frac{d^2E}{dT^2} = (t + 273) \times 2b$

$$E = \frac{Q}{C} + IR$$

$q = eV$
$\Rightarrow V \propto \frac{q}{C}$

$$\therefore E = \frac{Q}{C} + R\frac{dQ}{dt} \quad \text{——— (1)}$$

If, $Q = Q_0$ maximum charge at that instant,

$$I = \frac{dQ}{dt} = 0$$

∴ The P.D across the capacitor P is $E = \frac{Q_0}{C}$

From equation (1),

$$\frac{Q_0}{C} = \frac{Q}{C} + R\frac{dQ}{dt}$$

$$\Rightarrow \frac{Q_0}{C} - \frac{Q}{C} = R\frac{dQ}{dt}$$

$$\Rightarrow \frac{Q_0 - Q}{C} = R\frac{dQ}{dt}$$

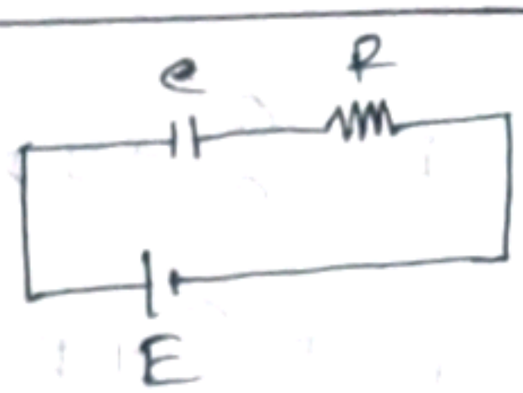
~~$\Rightarrow Q_0 Q = RC$~~

$$\Rightarrow \frac{dQ}{Q_0 - Q} = \frac{dt}{RC}$$

Integrating both side, $\int \frac{dQ}{Q_0 - Q} = \frac{1}{RC}\int dt$

$$\Rightarrow -\log(Q_0 - Q) = \frac{1}{RC}t + c \quad \text{——— (11)}$$

When, $t = 0$, $Q = 0$ then, $-\log Q_0 = c$



The emf, $E = \frac{Q}{C} + RI$

$$E = \frac{Q}{C} + R\frac{dQ}{dt} \quad \text{---(1)}$$

When, $Q = Q_0$ maximum charges, at that instant,

$$I = \frac{dQ}{dt} = 0$$

$$\therefore E = \frac{Q_0}{C}$$

putting this value in equation (1), we get,

$$\frac{Q_0}{C} = \frac{Q}{C} + R\frac{dQ}{dt}$$

$$\Rightarrow \frac{Q_0}{C} - \frac{Q}{C} = R\frac{dQ}{dt}$$

$$\Rightarrow \frac{Q_0 - Q}{C} = R\frac{dQ}{dt}$$

$$\Rightarrow \frac{dQ}{Q_0 - Q} = \frac{1}{RC} \cdot dt$$

Integrating both sides, we get,

$$\int \frac{dQ}{Q_0 - Q} = \frac{1}{RC}\int dt$$

$$\Rightarrow -\log(Q_0 - Q) = \frac{t}{RC} + c \quad \text{---(ii)}$$

when $t=0$, $q=0$. then

$$-\log q_0 = c$$

putting this value in equation (4), we get,

$$-\log (q_0 - q) = \frac{t}{RC} - \log q_0$$

$$\Rightarrow \log (q_0 - q) - \log q_0 = -\frac{t}{RC}$$

$$\Rightarrow \log \frac{q_0 - q}{q_0} = -\frac{t}{RC}$$

$$\Rightarrow \frac{q_0 - q}{q_0} = e^{-\frac{t}{RC}}$$

$$\Rightarrow 1 - \frac{q}{q_0} = e^{(-t/RC)}$$

$$\Rightarrow \frac{q}{q_0} = 1 - e^{(-t/RC)}$$

$$\therefore q = q_0 \left\{ 1 - e^{(-t/RC)} \right\}$$

(Hence showed)

Time constant, τ

$$RC = \frac{V}{I} \times \frac{q}{V} = \frac{qt}{q} = t$$

Now, let us put $t = RC = \tau$ in the equation of R-C circuit, we get,

$$q = CE \left\{ 1 - e^{(-\frac{t}{t})} \right\} = CE \cdot \left(1 - e^{-1} \right) = CE \left(1 - \frac{1}{e} \right)$$

$$= CE \times 0.63$$

$$= 0.63\, q_0$$

03.)

Given that,

$$E = 19.1t - 0.0152t^2 \quad \text{---(i)}$$

We know,

$$E = at + bt^2 \quad \text{---(ii)}$$

comparing (i) and (ii) ⇒

$$a = 19.1 \,, b = -0.015$$

Again, $E = at + bt^2$

$$\Rightarrow \frac{dE}{dt} = a + 2bt$$

$$\Rightarrow \frac{d^2E}{dt^2} = 2b$$

Now, T Neutral temperature,

$$T_n = \left(273 - \frac{a}{2b}\right) K$$

$$\Rightarrow T_n = \left(-\frac{a}{2b}\right)^\circ C$$

$$= \left(\frac{19.1}{2 \times 0.015}\right)^\circ C$$

$$\therefore T_n = 636.67^\circ C$$

Again, Temperature of inversion,

$$T_i = 2T_n$$

$$= 2 \times 636.67$$

$$\therefore T_i = 1273.33^\circ C$$

At $t = 30^\circ C$,

The peltier co-efficient,

$$\pi = T \frac{dE}{dT}$$

$$= 30 \times (a + 2bT)$$

$$= 30 \times (19.1 - 2 \times 0.015 \times 30)$$

$$= 18.2 \times 30$$

$$= 546^\circ C$$

The Thomson co-efficient,

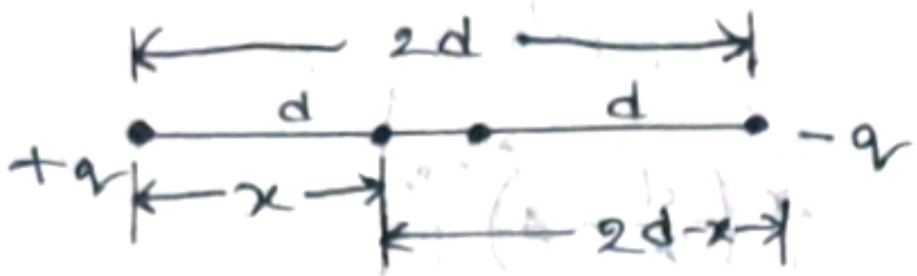
$$\sigma = T \frac{d^2E}{dT^2}$$

$$= 30 \times 2b$$

$$= 30 \times 2 \times (-0.015)$$

$$= -0.9^\circ C$$

☞ Deorive an expression for the electric field at a point on the axial line due to an electric dipole.



$\therefore r_{(+)} = d - x$

$r_{(-)} = [d - (d - x)] \; d$

$\therefore r_{(+)} = x$

$r_{(-)} = 2d - x$

$\therefore \quad E = E_{(+)} - E_{(-)}$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r_{(+)}^2} - \frac{1}{4\pi\epsilon_0} \frac{q}{r_{(-)}^2}$$

$$= \frac{1}{4\pi\epsilon_0} \left\{ \frac{q}{x^2} - \frac{q}{(2d-x)^2} \right\}$$

$$= \frac{q}{4\pi\epsilon_0} \left\{ \frac{1}{x^2} - \frac{1}{(2d-x)^2} \right\}$$

$$= \frac{q}{4\pi\epsilon_0} \left\{ \frac{1}{x^2} - \frac{1}{x^2\left(\frac{4d^2}{x^2} - \frac{4d}{x} + 1\right)} \right\}$$

$$= \frac{q}{4\pi\epsilon_0 x^2} \left\{ 1 - \frac{1}{\left(1 - \frac{4d}{x} + \frac{4d^2}{x^2}\right)} \right\}$$

$$= \frac{q}{4\pi\epsilon_0}\left\{\frac{1}{x^2} - \frac{1}{(2d-x)^2}\right\}$$

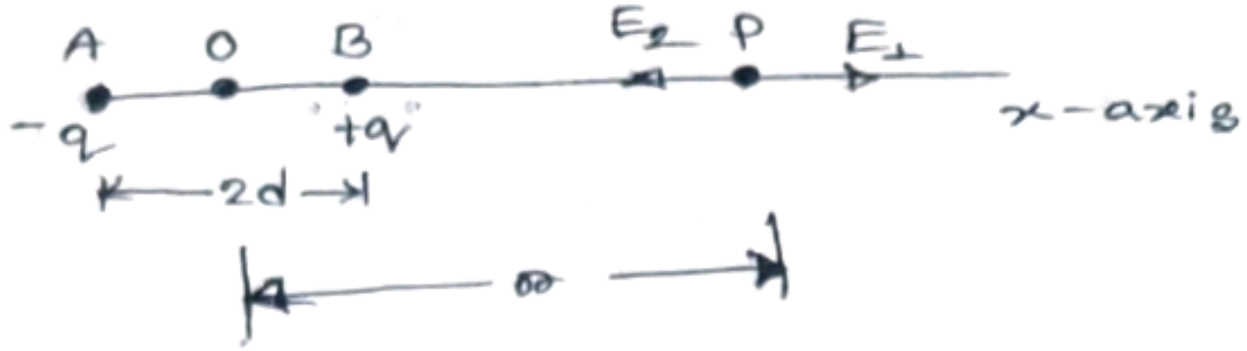
$$= \frac{q}{4\pi\epsilon_0}\left\{\frac{1}{x^2} - \frac{1}{\left\{x\left(\frac{2d}{x}-1\right)\right\}^2}\right\}$$

$$= \frac{q}{4\pi\epsilon_0 x^2}\left\{1 - \frac{1}{\left(\frac{2d}{x}-1\right)^2}\right\}$$

$$= \frac{q}{4\pi\epsilon_0 x^2}\left\{1 - \left(\frac{2d}{x}-1\right)^{-2}\right\}$$

2

☞ Derive an expression for the electronic field at a point on the axial line due to an electric dipole.



$$E_1 = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{(r-d)^2} \quad [\text{along } BP]$$

$$E_2 = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{(r+d)^2} \quad [\text{along } AP]$$

Now, $E = E_1 - E_2$

$$= \frac{q}{4\pi\epsilon_0}\left\{\frac{1}{(r-d)^2} - \frac{1}{(r+d)^2}\right\}$$

$$= \frac{q}{4\pi\epsilon_0}\left\{\frac{4rd}{(r^2-d^2)^2}\right\}$$

If the point P is far from the dipole, then $d < r$.

$$\therefore E = \frac{q}{4\pi\epsilon_0} \cdot \frac{4rd}{r^4} = \frac{q}{4\pi\epsilon_0} \cdot \frac{4d}{r^3}$$

$\therefore$ Electric field, $E = \frac{1}{4\pi\epsilon_0} \cdot \frac{2p}{r^3}$ $\left\{\because \text{Electric dipole moment, } p = q \times 2d\right.$

Q. A dipole is an external electric field or Potential Energy of a dipole in a uniform Electric field.

Sol:

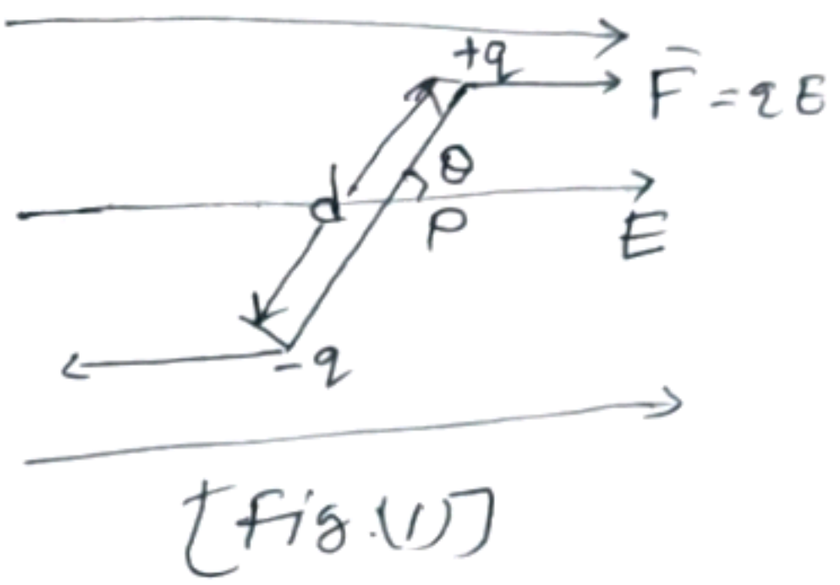
Consider a dipole of moment P makes an angle θ with a uniform external Electric field $\bar{E}$ [fig:(i)], the two forces F and -F acting on the two charges are equal but opposite. so that the net force on this dipole is zero but there is a net torque about an axis passing through the centre O. the magnitude of the torque is

$$\bar{\tau} = Fd \sin\theta \quad \text{——(i)}$$

but $F = qE$ ——(ii)

$$\therefore \bar{\tau} = dqE \sin\theta$$

or $$\bar{\tau} = PE \sin\theta \quad \text{——(iii)}$$



[fig:(i)]

The torque can be written in vector form as

$$\vec{\tau} = \bar{P} \times \bar{E}$$

suppose we choose the Potential energy of a dipole in an external field to be zero when $\theta = 90^\circ$. Let W be the work done by an external agent to turn the dipole from its reference orientation to angle θ then

$$W = \int_{90^\circ}^{\theta} \tau \, d\theta = \int_{90^\circ}^{\theta} PE \sin\theta \, d\theta = PE \int_{90^\circ}^{\theta} \sin\theta \, d\theta$$

$$\text{or } W = PE \left[-\cos\theta \right]_{90^\circ}^{\theta} = -PE \left(\cos\theta - \cos 90^\circ \right)$$

$$\therefore W = -PE \cos\theta$$

This work done is stored as potential energy in the dipole.

$$\therefore U(\theta) = -PE \cos\theta = -\vec{P} \cdot \vec{E}$$

* Gaussian surface: under necessary condition gauss law is applying for uniform, symetrical and charge distributional surface. this surface is called gaussian surface.

characteristics of gaussian surface:

1. the surface is hypothetical, not real
2. the surface may be circular, spherical or any other shape.
3. this surface might be enclosed
4. this surface might be uniformed and symmetric
5. charge is uniformly distributed through out the surface.